F = ma & Vector Notation

Exam code: 9MA0

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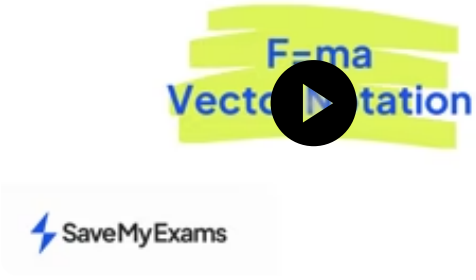
Written by: Paul
Updated on 16 February 2026



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F = ma & vector notation

How is F = ma used with vectors?

The **resultant force** (***F***) and **acceleration** (**a**) are vectors

- For **forces** and **motion** in **two dimensions**, ***F*** **N** and **a** m s^{-2} will be made up of two components – a **horizontal** (*x*-) **component** and a **vertical** (*y*-) **component**
- **Displacement**, **velocity** **and** **weight** are also vector quantities
- Time and mass are scalar quantities
- Vectors appear in **bold(**non-italic**)** font in textbooks, on exam papers, etc(i.e. **F**, **a**) but in handwriting should be underlined (i.e. F, a)

What notation is used for forces as vectors?

- **All** vectors are written either as **column vectors** or in **i–j format**
- As a **column vector F** = *m***a** would look like

$$\begin{pmatrix} F_x \\ F_y \end{pmatrix} = m\begin{pmatrix} a_x \\ a_y \end{pmatrix}$$

- In **i–j** notation **F = ma** would look like

$$F_x\mathbf{i} + F_y\mathbf{j} = m(a_x\mathbf{i} + a_y\mathbf{j})$$

Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

information given in the question – any vector quantities will be given as a **column vector** or written in **i-j notation**
- Remember ***F = ma*** is used when **motion** is involved – equations may come from '*suvat*' (if the acceleration is constant), or using N2L directly; look for (**resultant**) **force**, **mass** and **acceleration** being involved
- Use *F= ma*(N2L in 1D) or an appropriate '*suvat*' (in 1D) equation to set up and solve **separate** equations for both the **horizontal** (*x*-) and **vertical** (*y* -) components.

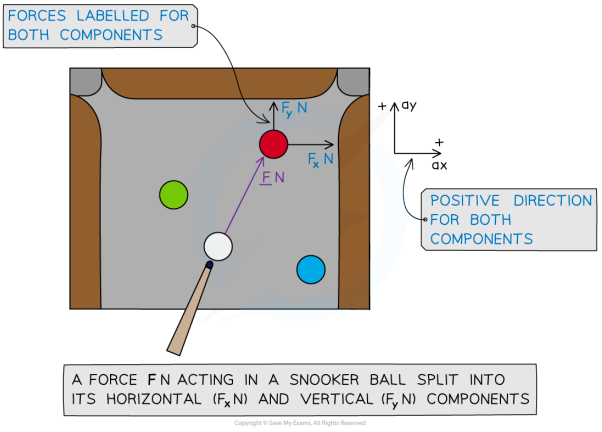
How is weight represented as a vector?

- Weight is a **force**, so it is a vector quantity
  - **W** = m**g** N where **g** m s^{-2} is the **acceleration** due to **gravity**

- Weight always acts **vertically downwards** so it only acts in the **j**-direction

$$\mathbf{W} = -mg\mathbf{j}N \qquad (g \approx 9.8 \text{ m s}^{-2})$$

- Treating the two dimensions separately means weight only needs to be considered when looking at the vertical (*y* -) direction
- Most 2D/vector problems are based on a bird's-eye view – the two dimensions being left/right and forwards/backwards, so the up/down (third) dimension where weight would apply, is often not involved



A Force, F, acting on a snooker ball

How do I apply F = ma in problems involving vectors?

- Step 1. If necessary, **draw a diagram** and label all forces acting on the particle(s)
  - label the **i** and **j** directions and any other useful information.
  - If a diagram is given, add any missing information to it.

- Step 2. Taking each dimension/component at a time use **F = ma**
  - If there is more than one particle involved you may have to do this for each

may arise

Worked Example

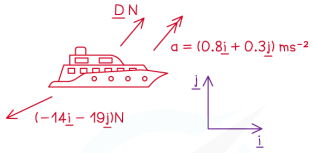
A yacht of mass 3.5 tonnes is sailing with constant acceleration $(0.8\mathbf{i} + 0.3\mathbf{j})\ \text{ms}^{-2}$.

The force from the engine of the yacht is **D** N. The yacht experiences a force from water resistance of $(-14\mathbf{i} - 19\mathbf{j})$ N.

Find the force **D** N acting on the yacht.

Answer:

Step 1: Sketch a diagram



Change units of mass: 3.5 tonnes = 3.5 × 1000 = 3500 kg

Step 2: A force is required, so let $D = (D_x i + D_y j)$ and use the 1D version of N2L ("F = ma") for both components

x: $D_x - 14 = 3500 \times 0.8$

y: $D_y - 19 = 3500 \times 0.3$

Step 3: Solve both equations and put back into vector form for the final answer

x: $D_x = 3500 \times 0.8 + 14 = 2814$

y: $D_y = 3500 \times 0.3 + 19 = 1069$

$\therefore D = (2814\mathbf{i} + 1069\mathbf{j})$ N

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Examiner Tips and Tricks

- If not given in the question, draw a diagram; label all forces and the positive direction for both components.
- Add to a diagram if given one, do not assume it is complete.
- Write a list of the quantities that are given in a question and another list of those you are asked to find. This will help you decide which equation(s) to use.
- A third list of the quantities you are not concerned with can help as these may be used to find intermediate results.

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